

Robustness of Structural Mode Transfer in MAAT v1.4

Alternative Frozen Fits, Regularisation Tests, and Multi-Domain Null Validation

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Abstract

Paper 52 found an asymmetric two-domain transfer result: a SAT-trained connectivity mode transferred strongly to a quantum pointer-state benchmark, while the reverse direction failed. The dominant signal in that experiment was the connectivity support V . The present paper tests whether this V -dominance is robust or merely an artifact of the fitting rule and the two selected domains.

We extend the frozen-weight protocol to four internal benchmark domains, not to independent external datasets: SAT frustration, quantum pointer robustness, active structural significance, and a deliberately synthetic societal CCI toy ensemble. Six source-only fit rules are compared: equal weights, non-negative least squares, positive ridge regression, positive Lasso, covariance-response closure, and a V -only baseline. All weights are frozen on the source domain and transferred without target-domain retuning or sign changes.

The result is mixed and should be read as an internal robustness audit. For the original SAT $\rightarrow$ Quantum transfer, the V -dominated signal is stable across NNLS, ridge, Lasso, covariance-response closure, and the explicit V -only baseline, with Spearman correlations $\rho_S \simeq 0.598\text{--}0.603$. Thus the Paper 52 result is not merely an NNLS artifact. However, across all sources and fitting rules, V is the dominant sector in only about 58% of frozen architectures, and equal weights achieve the largest mean cross-domain rank correlation due to support-composite internal toy targets.

The conclusion is therefore deliberately modest: connectivity behaves as a robust transferable mode for SAT-origin transfer, but it is not a universal transfer law. Structural transfer appears source-, scale-, and target-dependent, and the present benchmark primarily demonstrates internal consistency rather than external cross-domain validity.

1 Purpose

Paper 52 introduced a frozen two-domain transfer protocol. It found that SAT-trained structure transferred well to quantum pointer-state robustness, but that the reverse transfer did not solve SAT hardness. The result suggested that low-frequency structural modes may transfer even when domain-specific defect geometry does not.

The main open question was whether the SAT $\rightarrow$ Quantum result was specific to one fitting rule. In Paper 52, the SAT source fit collapsed to

$$\lambda_V = 1. \tag{1}$$

This is scientifically interesting but also fragile: if only one fitting choice produces the result, the transfer may be a fitting artifact rather than a structural mode.

The present paper therefore asks:

Does the V -dominated transfer survive alternative fitting rules, regularisation schemes, and additional internal domains?

2 Frozen Transfer Protocol

Each domain provides the shared supports

$$H, \quad B, \quad S, \quad V, \quad R_{\text{rob}}. \quad (2)$$

Supports are converted into support-defects

$$D_a(X) = -\log(\epsilon + \Gamma_a(X)), \quad \Gamma_a \in (0, 1], \quad (3)$$

and a frozen source-domain score is defined as

$$S_{A \rightarrow B}(X) = -\sum_a \lambda_a^{(A)} D_a^{(B)}(X). \quad (4)$$

Higher score is predeclared to mean higher structural quality. No target-domain retuning, sign flipping, or metric switching is allowed.

3 Domains

Four internal benchmark domains are used:

Table 1: Domains used in Paper 53.

Domain	Samples	Quality convention
SAT-Frustration	550	$1 - \text{minmax}(\log N_{\text{DPLL}})$
Quantum-Pointer	3500	$\text{minmax}(y_{\text{pointer}})$
Active-Significance	6000	$\text{minmax}(R_{\text{sig}})$
Societal-CCI	5000	$\text{minmax}(\text{ASI}_{\text{soc}})$

The additional Active-Significance and Societal-CCI domains are useful for stress testing, but they are not independent external validation. Their target variables are support-composite toy observables, so strong transfer into these domains is easier to obtain than transfer into SAT. This limitation is important for interpreting the aggregate statistics below.

The Societal-CCI domain should be read especially cautiously. It is a deliberately synthetic social-system toy ensemble designed to test whether the support language can separate coherence, balance, controlled activity, connectivity, robustness, and constructive impact. It is not a real social dataset, not a political ranking, and not evidence for external transfer.

4 Fit Rules

Six source-only fit rules are compared:

- **equal**: fixed equal weights.
- **nls**: non-negative least squares against source cost.
- **ridge_pos**: positive ridge-style regression with clipping and normalisation.
- **lasso_pos**: positive sparse regression.
- **response_top20**: covariance-response closure using the top-quality source quintile.
- **v_only**: fixed connectivity-only baseline.

The **v_only** rule is not a fitted model. It is included as a diagnostic baseline: if fitted transfer behaves exactly like **v_only**, then the result should be interpreted as connectivity transfer rather than balanced multi-sector MAAT transfer.

5 Results

5.1 Fit-Rule Summary

Table 2: Cross-domain performance by fit rule.

Fit rule	Mean ρ_S	Median ρ_S	Positive fraction	Mean λ_V
equal	0.5156	0.6661	0.7500	0.2000
response_top20	0.4630	0.4851	0.7500	0.4221
ridge_pos	0.4605	0.4537	0.7500	0.4014
nnls	0.4547	0.4644	0.7500	0.5188
v_only	0.4526	0.4085	1.0000	1.0000
lasso_pos	0.4139	0.3260	0.8333	0.6508

Equal weights have the highest mean cross-domain Spearman correlation. This should not be overinterpreted as an optimal universal rule. The active-significance and societal domains are constructed from support combinations, so broad support averaging is naturally advantaged. Nevertheless, the equal-weight result is an important negative control: in the current partly support-composite domain set, simple broad averaging remains competitive with more sophisticated source-specific fitting.

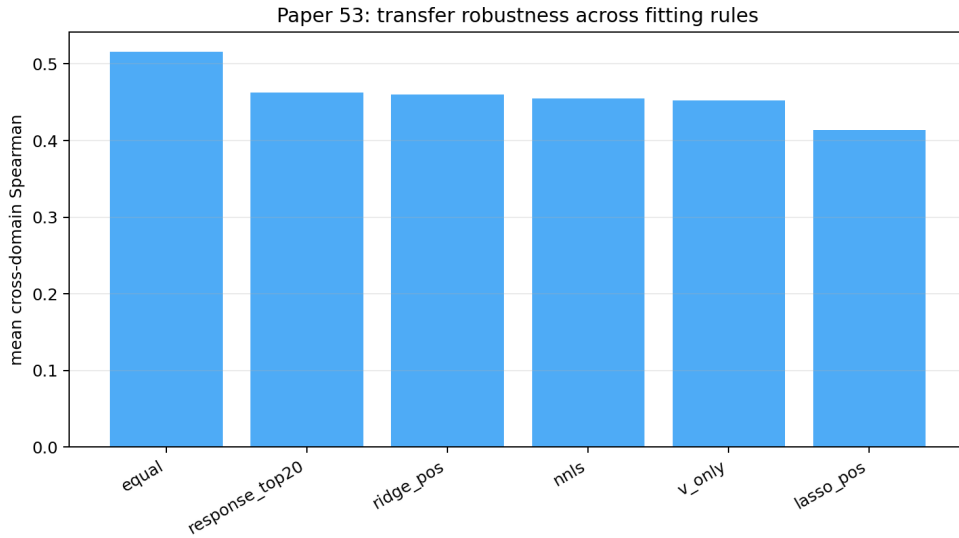


Figure 1: Mean cross-domain Spearman correlation by source-only fit rule.

5.2 The SAT $\rightarrow$ Quantum Test

The central robustness check is the original Paper 52 transfer:

Table 3: SAT $\rightarrow$ Quantum transfer under alternative fit rules.

Fit rule	ρ_S	Mutual information	λ_V
equal	0.3771	0.2043	0.2000
nnls	0.6034	0.3371	1.0000
ridge_pos	0.5981	0.3381	0.8394
lasso_pos	0.6034	0.3371	1.0000
response_top20	0.5975	0.3395	0.8264
v_only	0.6034	0.3371	1.0000

This is the strongest result of the paper. The SAT $\rightarrow$ Quantum transfer is not an NNLS artifact: it survives ridge regularisation, sparse positive regression, covariance-response closure, and the explicit V -only baseline.

5.3 V-Dominance Across Sources

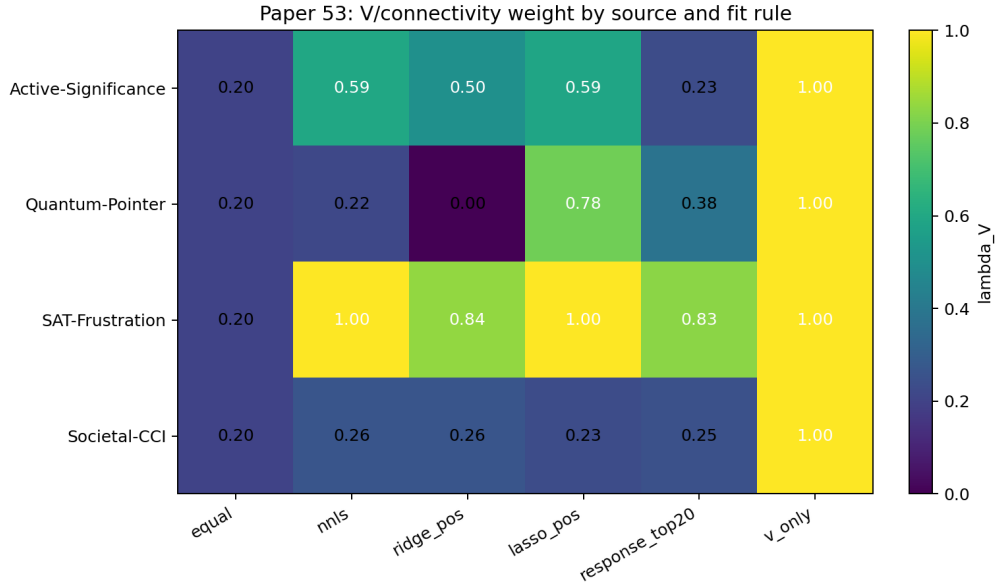


Figure 2: Connectivity weight λ_V by source domain and fit rule.

The heatmap shows that V -dominance is robust for SAT as a source and often appears in the active-significance benchmark. It is not universal. Quantum and societal sources can select H , S , or mixed structures depending on the fit rule. Across all source/rule pairs, V is dominant in roughly 58% of frozen architectures.

Paper 53: frozen transfer matrices by fit rule

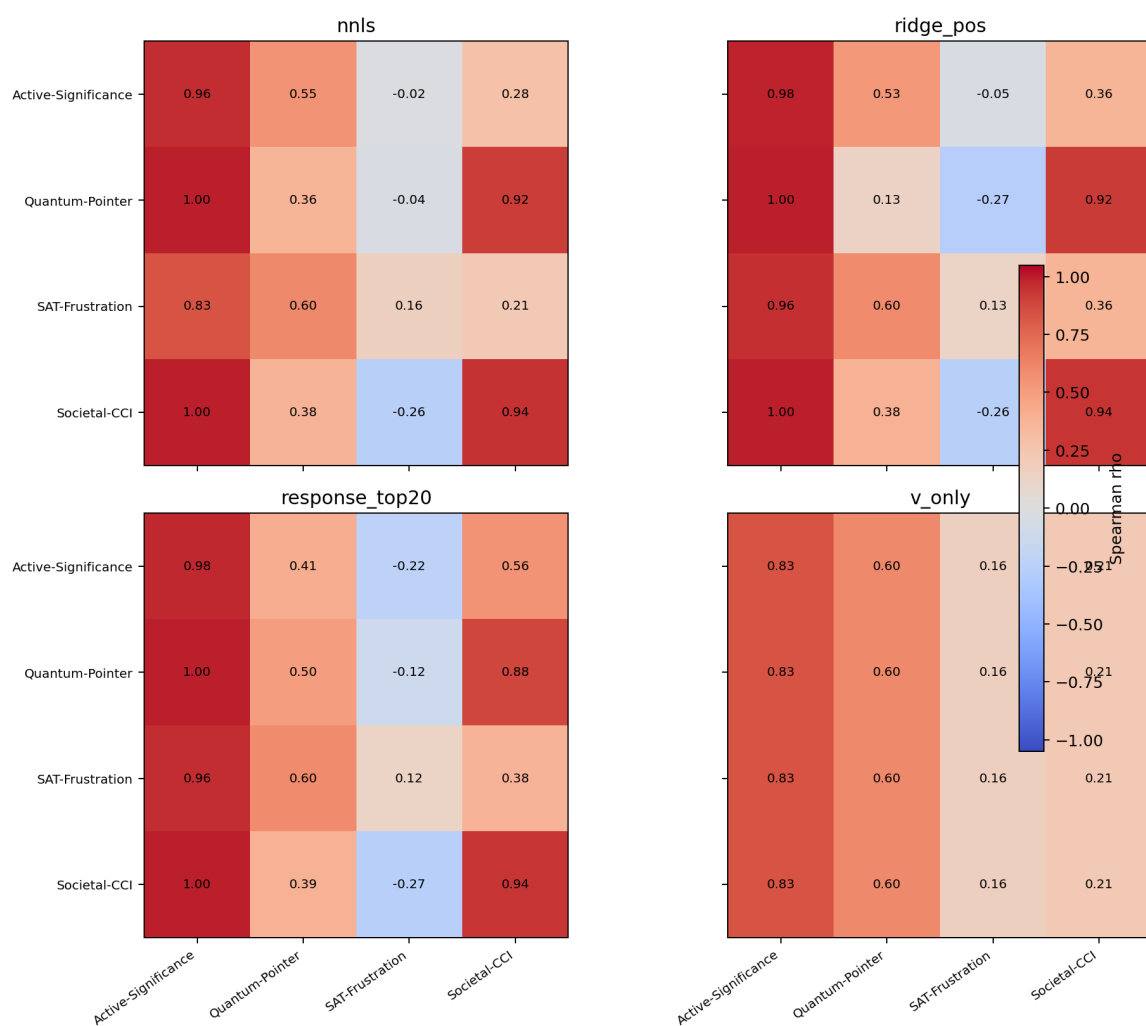


Figure 3: Frozen transfer matrices for four representative fit rules.

5.4 Shuffled-Defect Null Margins

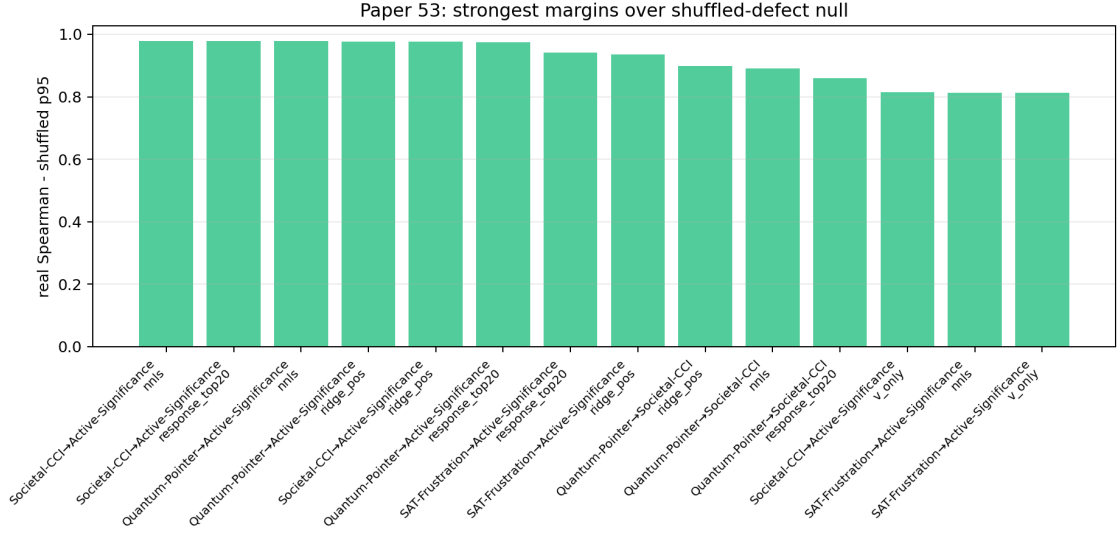


Figure 4: Largest margins over shuffled-defect nulls. Positive values indicate that the real frozen score exceeds the 95th percentile of shuffled-defect scores.

The shuffled-null comparison confirms that several strong transfers use sample-level support coherence rather than only marginal support distributions. However, the null analysis also reinforces the main caution: strong margins are concentrated in internal toy domains and should not be read as independent external validation.

6 Interpretation

Paper 53 supports a refined reading of Paper 52:

V is a robust transferable mode for SAT-origin transfer, but it is not a universal structural transfer law.

The result is consistent with the v1.4 mode interpretation, but does not prove it. Transfer should not be understood as a single compressed scalar crossing all domains. Instead, different supports behave like approximate structural modes. Some low-frequency modes, especially connectivity, can transfer between conceptually distant domains. Other structure remains local, high-frequency, and domain-specific. The results therefore resemble an effective mode hierarchy rather than a single transferable scalar.

The reverse SAT problem remains. Quantum-derived architectures do not robustly solve SAT hardness. Even when Lasso or V-only yields a weak positive transfer into SAT, the correlations remain far below the strong support-composite toy-domain results. This confirms the earlier lesson: SAT hardness depends on local frustration geometry and cannot be reduced to a simple global support score.

7 Limitations

- All domains are internal synthetic or derived repository artifacts.
- Active-Significance and Societal-CCI use support-composite quality targets, which makes broad support transfer easier than in independent external data.

- The SAT target is DPLL node count, not CDCL conflict count or industrial SAT runtime.
- Fit-rule hyperparameters are fixed for this benchmark and not fully tuned.
- The experiment tests robustness of mode transfer, not microscopic derivation of the sectors.
- The result should not be presented as universal cross-domain transfer.

8 Reproducibility

The experiment is stored in:

`experiments/structural_mode_transfer_paper53/`

Run from the repository root:

```
cd experiments/structural_mode_transfer_paper53
python3 structural_mode_transfer_robustness_paper53.py
```

The script writes:

- `paper53_transfer_results.csv`
- `paper53_frozen_weights.csv`
- `paper53_fit_rule_summary.csv`
- `paper53_shuffle_nulls.csv`
- `paper53_summary.json`

No new external dataset is introduced. The inputs are synthetic or derived outputs from earlier repository experiments.

9 Conclusion

Paper 53 tests whether the Paper 52 connectivity result survives a more adversarial robustness check. The answer is mixed but useful. For $\text{SAT} \rightarrow \text{Quantum}$, V -dominance is robust under multiple fitting rules and regularisation schemes. Across all domains, however, V is not universal, equal broad support averaging can outperform fitted modes on support-composite toy targets, and SAT remains resistant to simple incoming transfer.

The result is therefore consistent with a mode-based v1.4 interpretation while rejecting a simplistic scalar-universality reading: structural transfer is real in some low-frequency modes, but it is sector-dependent, scale-dependent, and domain-dependent. Paper 53 therefore shifts MAAT transfer claims from universality toward conditional structural portability.

References

- [1] E. T. Jaynes, “Information Theory and Statistical Mechanics,” *Physical Review* **106**, 620–630 (1957).
- [2] C. Krieg, “SAT Structural Hardness IIb: Frustration-Field Compression and the Limits of Scalar Robustness,” preprint (2026).
- [3] C. Krieg, “Multi-Scale Defect-Field Structural Selection,” preprint (2026).
- [4] C. Krieg, “Quantum Pointer-State Selection in a Non-Commuting Lindblad Benchmark,” preprint (2026).
- [5] C. Krieg, “Cross-Domain Transfer of Structural Selection,” preprint (2026).